Global Facilities - Design & Construction Standard - I&C - TGMS

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1.0 Purpose

This Standard Specification defines the minimum requirements for design, supply, packing and delivery of Toxic and Flammable Gas Monitoring System to greenfield project, including installation, cabling, factory & site testing, and commissioning.

2.0 Scope

2.1 Site(s) impacted

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

2.2 Vendor Scope

2.2.1 The following are in the scope of TGMS vendor/contractor:

a. System sizing, control panel design
b. Preparation of the layout drawings.
c. I/O list
d. Cable schedules complete with all details such as Type, Size, Length, quantity of all cables.
e. Supply and install all necessary hardware, software, system network, cables, alarm devices and field instrumentation (gas detector) to build the TGMS in accordance with this specification.
f. Provide all the necessary services for designing, manufacturing, supply, delivery, installation, testing, commissioning, and handover.

g. TGMS functional specification for Micron approval.
h. Provision of Clean room / Hazardous area type devices (if applicable).
i. Liaison with other system vendors for necessary interfacing.
j. Program Software.
k. Operation & Maintenance Spares.
l. As-Built Drawings.
m. Authority Submissions, sign-off, and completion of Certificate of Readiness.
n. Operation & Maintenance Manual.
o. Safety Integrated Level (SIL) calculations if applicable.
p. Inspection and Test Plan (ITP).
q. Testing and Commissioning.
r. System Training
s. 24/7 Monitoring of system at control room from system commissioning to handover of the system to Micron.

2.2.2 The TGMS shall be designed and installed to operate under the project site conditions.

2.2.3 The equipment and software proposed by the vendor shall be currently in manufacturer. No custom products shall be allowed. All products shall be supported by manufacturer for a minimum of ten (10) years from date of acceptance, including spare parts, board repairs and software revisions.

2.2.4 TGMS shall include any special tools required for installation, field testing, commissioning, operation and maintenance of the TGMS.

2.2.5 Where required, commissioning spare parts are to be provided during commissioning of the system & equipment by the TGMS vendor.

2.2.6 The TGMS vendor shall include as optional, recommended 2 years operational spares.

2.2.7 TGMS vendor shall ensure that all equipment and materials are designed and packed for transit to site without damage.

2.2.8 The TGMS vendor shall ensure a proper organization structure for the project development and management. The vendor shall submit list of persons and resume of everyone who will perform project manager, engineer, and construction management functions in the project.

2.2.9 The TGMS vendor shall deliver products or services as per requirement specification and shall have an appropriate quality management system.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
AC	Alternating Current
BIC	Building Isolation Cabinets
BKM	Best Known Method
CPU	Central Processing Unit
DC	Direct Current
EHS	Environmental, Health, and Safety
EMO	Emergency Off
ERT	Emergency Response Team
E-STOP	Emergency Stop
EVB	Exhausted Valve Box
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
FDB	Field Distribution Box
FMEA	Failure Mode and Effects Analysis
GDB	Gas Distribution Box
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
GPS	Global Positioning System
HAZOP	Hazard and Operability
HMI	Human Machine Interface
I/O	Input/Output
IDLH	Immediately Dangerous to Life or Health
MOC	Management of Change
MTBF	Mean Time Between Failures
P&ID	Piping & Instrumentation Diagram
PLC	Programmable Logic Controller

Term/Acronym	Definition / Description
PSM	Process Safety Management
SCADA	Supervisory Control and Data Acquisition
SIL	Safety Integrity Level
TGMS	Toxic Gas Monitoring System
TLV	Threshold Limit Value
VMB	Valve Manifold Box

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Incorporate all relevant BKM, lesson learned and emerging technologies in System Standard document. • Perform periodic document review and update System Standard.
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard. • Support the Site teams by providing knowledge and technical expertise.
Site Project Construction Teams	• Review and provide technical feedback on System Standard. • Incorporate System Standard into Project Engineering, Design, and Construction. • Document project deviations, exceptions, and exclusions from System Standard. • Share new lessons learned to GFTT Lead, to determine if it is BKM that can be captured in future revision of System Standard.
Site Facilities Teams	• Use System Standard as technical reference.

5.0 Applicable Codes and Standards

The following industry, association, codes and standards shall be followed as applicable to the design, fabrication, assembly, and testing of all equipment furnished under this specification. Deviations from this must be agreed formally by Micron.

a. ANSI/IEEE Standard
b. Instrument Society of America (ISA)
c. IEC 61511
d. IEC 61508
e. Micron Corporate EHS Toxic Gas Monitoring Standard
f. Micron Corporate EHS Toxic and Flammable Gas Monitoring Matrix
g. Global EHS – EMO Button Guideline.

Information in the Corporate EHS *Toxic and Flammable Gas Monitoring Matrix* include, but is not limited to, type of gas, monitoring requirements, alarm limits, evacuation levels, and toxicity.

Electrical and instrument equipment and materials for installation should comply with hazardous level requirement. Refer to NFPA-496.

6.0 General Concept and Requirement

6.1 This Specification defines Vendor scope for supply and install of the TGMS and relevant field instrumentation for Micron project. It is intended that the vendor shall check, evaluate and further develop this specification as necessary to suit the detail engineering.

6.2 Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall refer to Appendix A in this specification.

6.3 Main purpose of TGMS is to monitor the level / presence of different gasses against define limits and take appropriate action when these limits are exceeded such as activate alarm, dry contact interface provision, shutdown equipment, etc.

6.4 Exhausted enclosures containing toxic or flammable gases which are specified in TGM matrix shall be monitored. This includes, but is not limited to, VMBs, BIC, GDBs, FDBs, EVBs, gas cabinets, pump packages, pollution abatement rooms, and chemical rooms.

6.5 To design TGMS as safety instrumented system, safety life-cycle plan in accordance with IEC-61511 shall be implemented. Refer to **Figure 1** for the phases in safety life-cycle. In the beginning of the TGMS design, Hazard and risk assessment shall be performed with FMEA or HAZOP techniques. The Hazard and risk analysis shall define SIL requirement. It is best performed in a team of relevant departments/organizations including, not limited to: vendor, operator, control design engineer, process engineer, EHS/PSM, and project management.

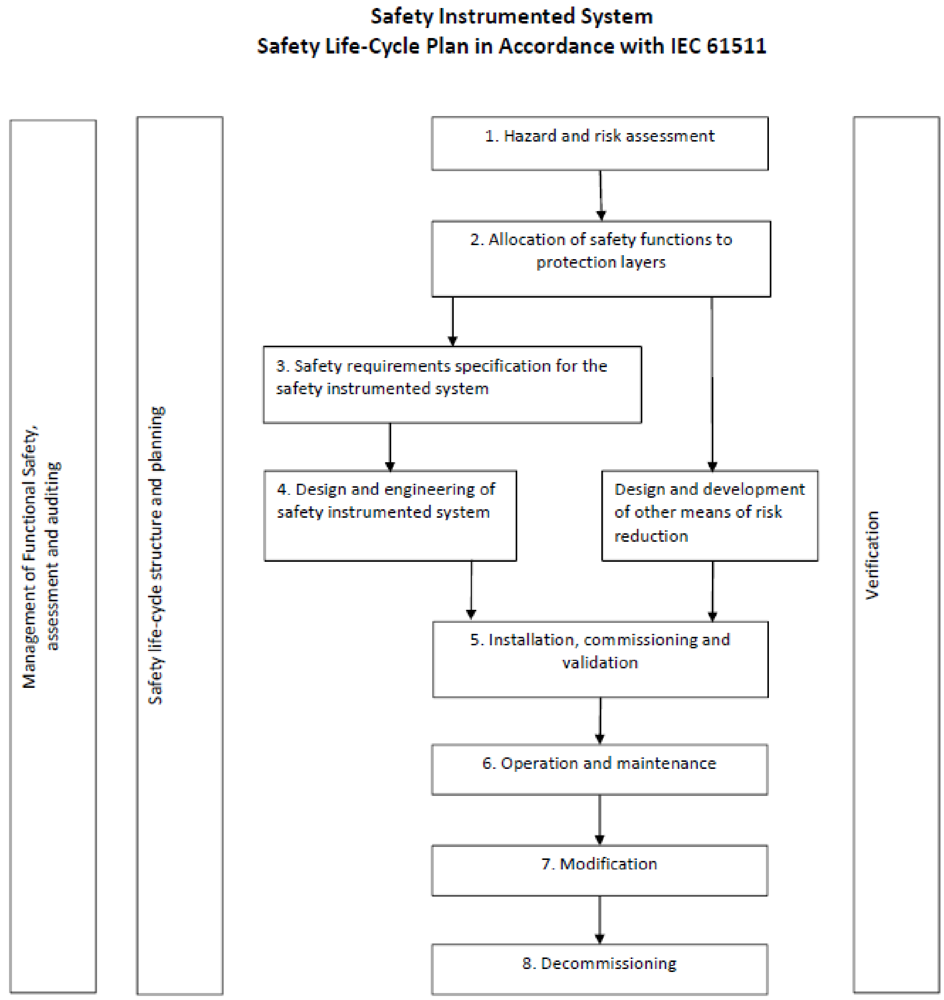
6.6 When determining additional area monitoring requirements, an analysis of each toxic gas system shall be performed, which incorporates the following:

6.6.1 Analysis shall be performed in coordination with gas/chemical and controls engineering, and EHS and PSM.

6.6.2 Analysis shall determine the placement of sensors as a function of, but not limited to, area air flow characteristics, mechanical piping and fittings utilized, piping failure points, sensor capabilities and protection of personnel.

In general, the design review, and approval shall be performed in coordination with gas/chemical, control, and EHS/PSM representatives.

Figure 1: Safety Life-Cycle Plan



6.7 The monitoring system capacity and performance requirements:

6.7.1 Processors and memories used for continuous monitoring and interlocks should appear used less than 50% of their capacity, to allow future development.

6.7.2 Network performance or loading shall be maximum 30%.

6.7.3 The time between status change in the field and alarm displayed in the SCADA will be less than 2 seconds.

6.7.4 Analog measurement alarms will be generated within 2 seconds from alarm condition occurrence. Normal value updates in the display will be within 3 seconds after change.

7.0 Alarm Requirement

7.1 At a minimum, any gas monitoring system at a Micron site must have the capability of reacting to the three scenarios as defined by the investigation level and alarm level, specified in:

7.1.1 Global EHS – Toxic Gas Monitoring and Double Containment Standard

7.1.2 Global EHS Toxic and Flammable Gas Monitoring Matrix

7.2 At any Alarms shall be generated for the following purposes:

7.2.1 Early Detection – Any detectable reading above zero or background masking level.

7.2.2 Evacuation Alarm – Evacuates the breathing zone.

7.2.3 Shutdown and Alarm - Automatically shuts down gas and alarms breathing zone.

7.3 Alarms shall be provided to monitor the system in case of a fault. Fault alarms include, but are not limited to, Warning, Trouble, Low-Pressure, Low-Flow and Shutdown.

7.4 The following alarms shall be provided to notify maintenance, operations, and / or security personnel:

7.4.1 System Fault – Defined as any system abnormality or loss of functionality, including detector fault and loss of power or communications. It shall notify FMCS control room of maintenance and equipment health.

7.4.2 Early Detection – Defined as any detectable reading above zero or background masking level. It shall notify FMCS control room of maintenance and equipment health.

7.4.3 Evacuation Alarm. It shall notify FMCS control room and security control.

7.4.4 Shutdown Alarm. It shall notify FMCS control room, security control room, and tool owners.

7.5 Blue strobes shall be used to notify occupants that a gas has been sensed at an evacuation level within their breathing zone or area. The evacuation procedure shall be as per site response plan.

7.6 Systems shall be designed to be fail-safe. Valves, relays, and instruments shall fail to a safe operational position until control can safely be re-established, and the safety of personnel is adequately established.

7.7 A system assessment shall be performed to determine whether any area monitoring points are required. See the Micron Safety Standard for Toxic Gas Monitoring for monitoring strategies based on system assessment.

7.8 Horns and Blue lights must be placed as specified in Corporate EHS Toxic Gas Monitoring and Double Containment Standard, (Doc ID: 2W4373RQWREN-1568922467-11).

7.8.1 Strobes - Individual wall and ceiling strobe units shall incorporate the following features:

i. Power Requirements: 24VDC
ii. Bezel Color: Blue
iii. Lens Color: Blue
iv. Manufacturer: Gentex Commander Series

7.8.2 Horn/Strobe Units – Horn/Strobe units shall incorporate the following features:

i. Power Requirements: 24 VDC
ii. Bezel Color: Blue, Ceiling mounted devices shall be painted.
iii. Lens Color: Blue
iv. Manufacturer: Gentex Commander Series, Wall and Ceiling mount.
v. For class 1 division 2 areas the horn/strobe shall be MEDC Cooper Industries,24 VDC, PN# SM87HXB/DB3. Device housings shall be painted Blue in color. Lens Color: Blue
vi. Minimum one Horn and alarm lamp are to be installed in one room of gas effected room. If two different events can activate Horn and alarm lamp, the operator must act for the worst-case scenario.
vii. Horn and Strobe-light of TGMS are recommended to be independent from Fire Safety system.

8.0 Communication and Network Infrastructure

8.1 Control network that communicates among equipment should be redundant.

8.2 Control network that communicates among PLC-PLC should be dedicated and redundant.

9.0 I/O List Requirement

The I/O list shall have at least, the following information to assign each I/O:

a. SCADA point tag
b. Building/Area
c. Equipment Type
d. Equipment Name/Tag
e. Equipment Description
f. Service
g. Process Name/Tag
h. Instrument/Device Type
i. Power supply requirement
j. Scope of instrument/device supply and install
k. P&ID / drawing / document reference.
l. Control Panel Tag/Name
m. Panel location
n. I/O Type (Digital Input, Digital Output, Analog Input, Analog Output, Pulse Input, High Level (OPC, Modbus, Profibus, etc), Others – to be specified)

10.0 Control Operation

10.1 The control shall refer to Global EHS – Toxic Gas Monitoring and Double Containment Standard, document number: 2W4373RQWREN-1568922467-11)

10.2 At the minimum, once a gas leakage has been detected, the detection system must be capable of:

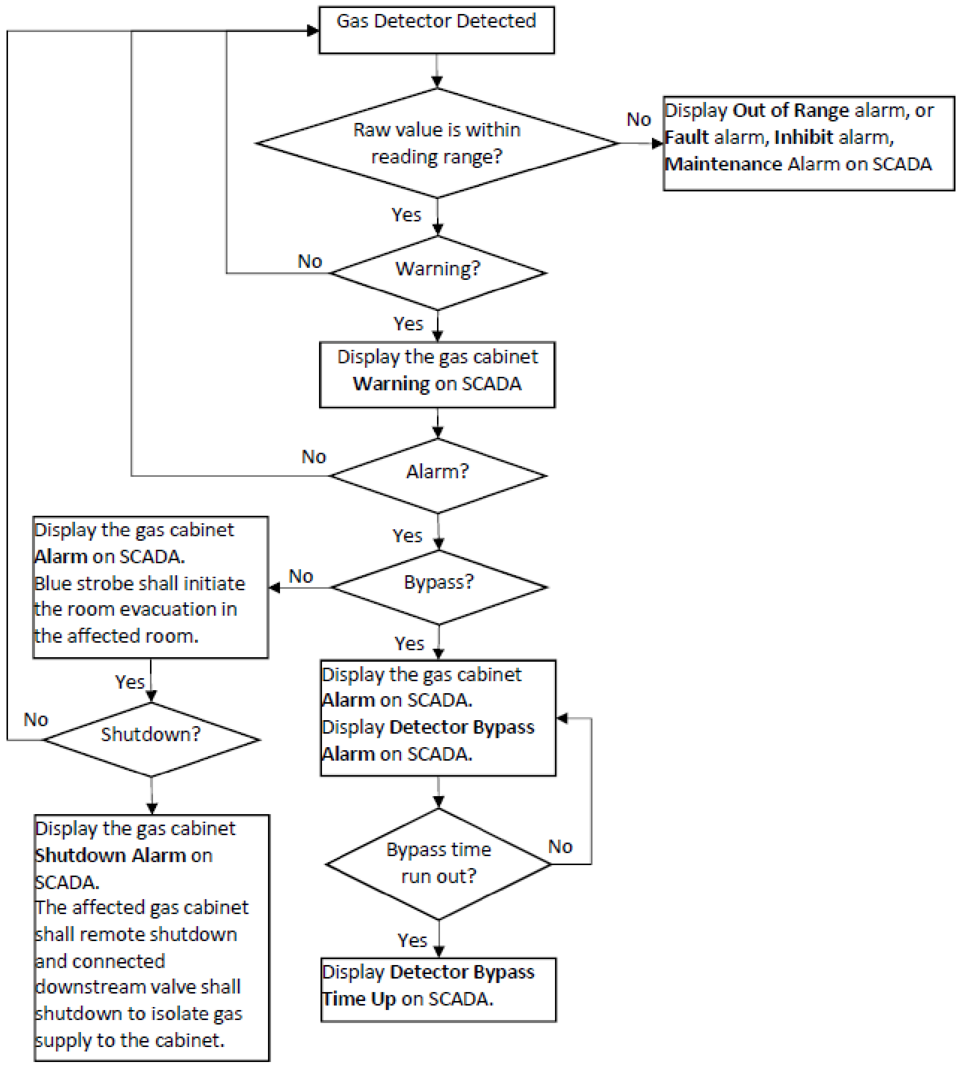
10.2.1 Activating a local audible evacuation alarm

10.2.2 Sending a notification signal to the central control station (SCADA)

10.2.3 Triggering auto shut-off of the detected gas supply.

10.3 The flow chart in **Figure 2** in general applies to gas detection control:

Figure 2: Gas Detection Control Flow Chart



10.4 To reset the alarm, first clear alarm conditions, i.e. clear all hazardous gases. After readings return to normal: (1) Reset gas cabinet and its gas supply cabinet via SCADA; (2) Reset Remote shutdown at local gas cabinet and the downstream valve shutdown.

10.5 Connection between controller and alarm devices shall be hardwired.

10.6 After an alarm has tripped an equipment, the alarm must be reset manually before the equipment is able to operate again.

10.7 Visual notification circuitry should use separate circuits but be integrated with the fire alarm system for notification of personnel. This apply to Natural Gas system.

10.8 Audible notification circuitry of TGMS should be integrated with the audible annunciation circuitry of the fire alarm system, to take advantage of sharing annunciation circuitry throughout the Fab areas.

10.9 The system shall operate in a fail-safe mode so that when the ability to control the gas supply has failed, the system will shutdown the supply until safe control can be restored.

10.10 The system shall operate in a fail-safe mode so that if a toxic gas sensor fails, the system will alarm and fail to a condition that ensures safety of the personnel in the area.

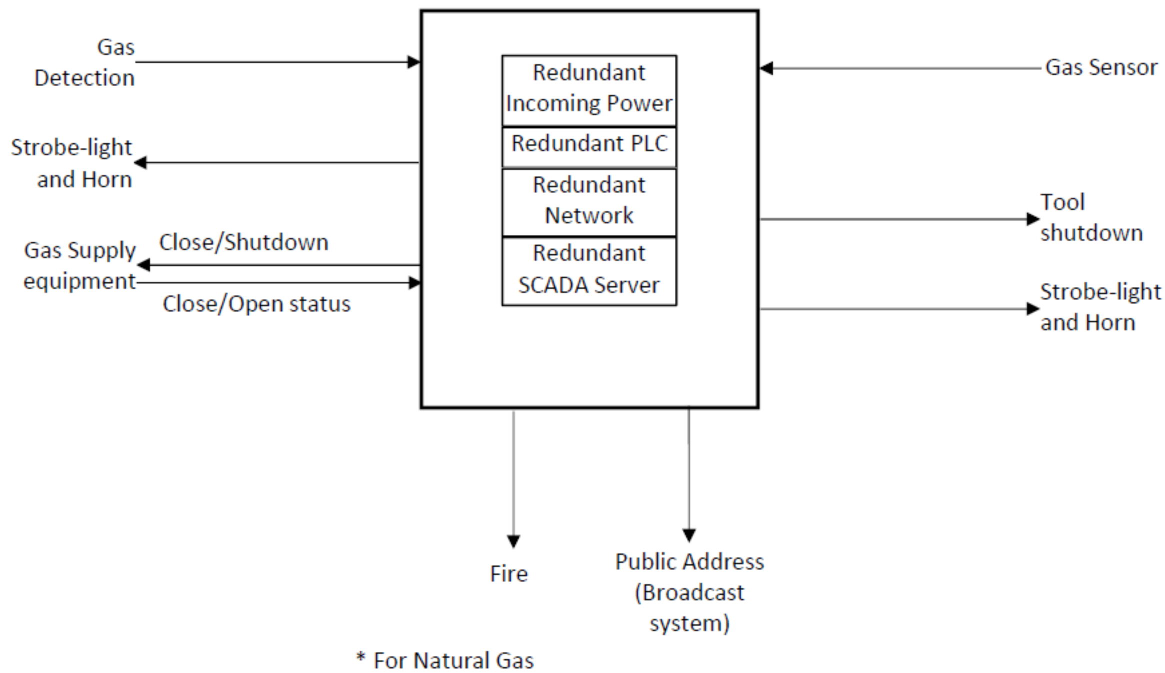
10.11 The evacuation control circuitry shall incorporate normally closed or supervised circuitry to ensure that the integrity of the circuit provides personnel with confidence in the circuits' ability to protect them if a gas is detected.

10.12 EMO shall be monitored by TGMS SCADA. If EMO is activated, the gas cabinet in the affected gas room shall shutdown, Blue Strobe shall activate in the affected gas room for evacuation, the affected Gas room EMO Activate alarm shall be displayed on SCADA, and the gas cabinets shutdown alarm shall be displayed on SCADA.

10.13 Vendor shall develop functional specification with test specification to meet this requirement.

10.14 If hazard and risk assessment define to have more than one sensor for an equipment/tool, the voting requirement needs to be reviewed by Control, Chemical/Gas and EHS/PSM representatives.

Figure 3: TGMS interface diagram



11.0 Hardware

11.1 PLC

11.1.1 The control unit shall be modular and microprocessor based and shall allow the continuous monitoring and the display of the operating status (on/off, alarm/failure) of every connected sensor.

11.1.2 The system shall utilize a PLC network to provide gas shutdown, evacuation, and alarm capabilities to gas equipment. Networks shall utilize distributed I/O for shutdown and evacuation circuitry, so that the I/O is located close to the gas distribution equipment to be shutdown to allow for the standardization of visual notification appliances.

11.1.3 Fault tolerant including redundant power supplies, CPUs, and network.

11.1.4 Conformance to current revisions of standards, but not limited to: ISO 9001, IEC 60664 and IEC 61000 (Isolation), IEC 60668 and IEC 60529 (Environmental), and IEC 60255.

11.1.5 On-line hardware, software, and communications diagnostics tools. The diagnostic messages of CPU, power module, communication, and I/O modules shall be sent to SCADA for monitoring and alarming. The status of each module and system communication status shall be monitored and displayed with its location in the SCADA.

11.1.6 Connectivity to SCADA field devices, HMI.

11.1.7 Use copper or optical fiber media or both for network communications to optimize PLC network speed.

11.1.8 Design network with segments and limited number of PLCs per segment to optimize communications speed.

11.1.9 Straightforward and efficient installation, calibration, and maintenance.

11.1.10 Provide local and factory vendor technical support.

11.1.11 Conformal coating for environmental protection.

11.1.12 Operate within a temperature range of 0 to 60 degrees Celsius and 5 to 95 percent of relative humidity (non-condensing) at 60 degrees C.

11.1.13 Minimum MTBF requirements: 250,000 hours for processors; 500,000 hours for communication modules; 1,000,000 hours for I/O modules.

11.1.14 Each controller shall be sized in accordance with the following requirements:

i. Additional I/O should be included to provide 20% extra spare points of each I/O type (AI, AO, DI, DO) for future expansion.
ii. 20% spare space shall be available on control rack to allow the addition of future modules.

11.1.15 Provide blank cover plates for spare PLC slots.

11.1.16 Provide PLC communication cables satisfying requirements of PLC manufacturer and approved by Micron. Install cable in accordance with installation requirement of the PLC manufacturer.

11.1.17 Time in PLC shall synchronize with FMCS network or GPS time.

11.2 I/O Interface

11.2.1 In every module I/O including spare will be fully wired.

11.2.2 Under normal operation, the relay is to be energized. Upon relay failure, the valve shall close.

11.2.3 Relay circuits shall be supervised or utilize a closed contact under normal operation.

11.2.4 Signal lines from redundant sensors shall connect to different I/O modules in different rack.

11.2.5 I/O max. points of each PLC hardware shall be 4000 points.

11.2.6 I/O max. points for each remote I/O station shall be 1000 points.

11.2.7 I/O Bypass circuit shall be provided for maintenance/troubleshooting. The bypass status must be monitored by TGMS.

11.3 SCADA

11.3.1 To define latest server hardware specification based on high-end server capability and IT requirement.

11.3.2 The server shall capacity to retain historical data for minimum 2 years.

11.4 SCADA Client

11.4.1 To define latest server hardware specification based on high-end capability and IT requirement.

11.5 Control Panel

11.5.1 Fabricate control enclosures 48 inches by 36 inches or smaller suitable for wall or a strut rack mounting.

11.5.2 Fabricate control enclosures larger than 48 inches by 36 inches as freestanding panels fitted with removable lifting lugs and legs minimum 12 inches high. Provide steel nuts and bolts with gaskets to fill holes after removal of lifting lugs.

11.5.3 Control Panels and Enclosures:

i. Standard size prefabricated enclosures manufactured by Hoffman or Rittal or approved equal.
ii. Enclosures will be NEMA 4X for wet, damp and corrosive locations; NEMA 12 for all other locations.
iii. Oil-tight latching mechanisms with keyed-alike locking handle. Tape keys to inside of doors.
iv. Gasketed overlapping front access doors, with heavy gauge hinges and three-point latch rods with rollers.
v. Internal LED panel light with safety cover, end cap, and convenient on/off switch. Provide one fixture for each 48 inches of panel width.
vi. 9 inches by 12 inches by 1-inch print pocket to store arrangement, schematic, and loop drawings.
vii. Minimum 14-gauge steel.
viii. Electrical power sockets that are isolated from the control wiring.
ix. Cooling package (e.g. fan exhaust, enclosure air conditioner coolers) sized for published heat rejection information for powered components. Incorporate integral filters and sealing gaskets into fans and exhaust vents without compromising the panel environmental rating.
x. Install enclosure heaters on enclosures that will not be located within climate-controlled spaces to prevent condensation and resulting corrosion.

11.5.4 All field wiring shall be bottom entry.

11.5.5 Provide wiring between terminal blocks and panel equipment.

11.5.6 Fasten components (e.g. terminal blocks) to interior surfaces of panels. Mount operator interface components on front access door/panel.

11.5.7 Install copper grounding bus bar and studs, and provide 10% (minimum of 3) spare taps and screws for future and field grounding use.

11.5.8 Provide ground cable connection over hinge between subpanel and door.

11.5.9 Provide separate wireways for AC and DC signals. Separate AC wiring from DC wiring. Maintain a 4-inch minimum spacing between parallel runs of power and control wiring and electronic instrument wiring.

11.5.10 Size wireways so that total cross-sectional area of installed wire will not exceed 20% of cross-sectional area of wireway. Locate wireways to eliminate interference with access to panel-mounted instruments or devices. Do not encroach on space allotted for future instruments and do not install terminal strips and splices in wireways.

11.5.11 Group wires to panel-mounted instruments and switches through wireways or cable ties.

11.5.12 Provide finger-safe terminals and components per IP-20.

11.5.13 Cut, punch, or drill panel cutouts and smoothly finished with rounded edges.

11.5.14 Provide one-piece back panels.

11.5.15 Mount components to the back panel using machine screws or bolts with tapped holes according to manufacturer’s recommendations. Provide lock washers with bolts and screws.

11.5.16 Mount components such that:

i. Adjustments can be made without the removal of instruments or interruption of service to adjacent components.
ii. Their installation will not be affected by, or interfere with, the removal of panel instruments.
iii. Components can be removed from the front without removal of the back panel.

11.5.17 Support door-mounted equipment such that no appreciable or noticeable sag occurs and no excessive wear on the hinges or wiring occurs.

11.5.18 Provide lockout/tagout capable main power switch with circuit breaker or fused disconnected and terminal block for termination to each AC power (120 or 240V) feeder circuits. Provide a power terminal block for termination of the wires.

11.5.19 Provide power distribution component layout and wiring in accordance with the Drawings. Provide individually protected Vac and 24Vdc circuits for instruments as required plus 20% for future.

11.5.20 Provide a circuit breaker panel with circuit breakers for each specific power source to accommodate panel lighting, power supplies, instruments, I/O modules, and processors.

11.5.21 Provide sufficient space to allow AC power cables to run directly from the power block to the circuit breakers.

11.5.22 Terminate internal power wiring so that an interruption at any one point will not disrupt power to remaining components.

11.5.23 Wiring for Front of Panel Devices:

i. Provide hinge wiring wrapped with plastic cable stays. Secure hinged wiring at each end so that any bending or twisting will be around the longitudinal axis of the wire. Protect the band area with a sleeve.
ii. Attached wiring to interior side of panel door to house wires for panel front-mounted devices.

11.5.24 Terminate interconnecting wires, including spares, between panel-mounted equipment and external equipment at numbered terminal blocks. Multi-conductor cable is not allowed except for I/O module wiring harnesses.

11.5.25 Arrange wiring neatly, cut to proper length, and remove surplus wire. Provide abrasion protection for wire bundles, which pass through holes or across edges of sheet metal. Spliced or taped wiring, wire nuts, and butt splices are not acceptable.

11.5.26 Wire Labels:

i. Preprinted, slide-on type or heat shrink cover Brady type, which may be wrapped around wire.
ii. Affix wire labels at both ends of wire.
iii. Cloth or adhesive tape is not acceptable.

11.5.27 Install wiring in such a way to provide protection from inadvertent grounding and cross connection.

11.5.28 Service Legends and Nameplates: Engraved, rigid, laminated plastic type with adhesive back; White with black letters unless otherwise noted.

11.5.29 Firmly affix interior front of panel nameplates to the panel without compromise enclosure rating.

11.5.30 Firmly affix interior nameplates next to the device with industrial grade adhesive.

11.5.31 Comply with grounding requirement as per NFPA 70 or equivalent.

11.6 Gas Detector

11.6.1 Refer to Global EHS – Toxic Gas and Double Containment Standard, that Gas Detecting Monitoring Criteria:

Micron's Toxic and Flammable Gas Monitoring Matrix has been developed utilizing the International Fire Code (IFC) and International Building Code (IBC) requirements for gas monitoring. These requirements state that a continuous gas detection system must be provided for Hazardous Production Material (HPM) gases when the physiological warning properties of the gas are at a higher level than the accepted OEL for the gas, and for flammable gases. Local legal requirements for gasses requiring detection must be evaluated on a site by site basis.

11.6.2 Recommended manufacturers are Honeywell, DOD, or approved equal.

11.6.3 All field-mounted devices shall have nameplates. All components shall be provided with stainless steel nameplate showing manufacturer; tag no.; serial no.; model no. as minimum.

11.6.4 All gas detectors shall be pre-calibrated and documented with a calibration certificate by the vendor.

11.6.5 Detectors are installed in such a way that they are well accessible for calibration and maintenance work. The cable length must be selected in such a way that the detector can be calibrated without disconnecting it from the instrument.

11.6.6 Hazardous gas detection equipment shall provide detection of hazardous gasses at or below the Permissible Exposure Level (PEL) / Threshold Limit Value (TLV).

11.6.7 Flammable gas detection equipment shall provide detection at or below the 10% Lower Explosion Limit (LEL) for hydrogen and solvent fumes.

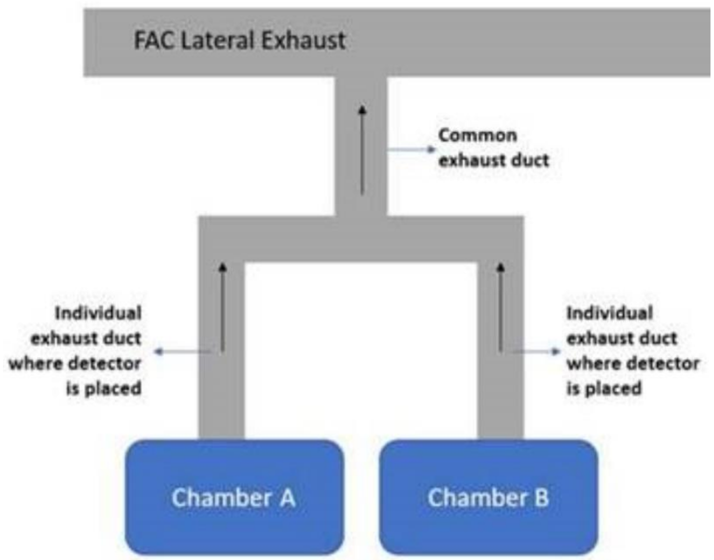
11.6.8 Compression fittings shall be used on sample connections to exhaust ducts. Sample and exhaust tubes shall be FEP Teflon.

11.6.9 All new tubing shall be tested for integrity and functionality after it is installed.

11.6.10 Location of sampling equipment shall be such that the manufacturers' maximum required distance for sampling tube length is not exceeded.

11.6.11 When monitoring an enclosed exhausted piece of equipment, the monitoring point should normally be in the ductwork of the exhaust. It is important to verify that the monitoring point is situated in a way that allows it to properly detect the target gas without interruption, such as air turbulence. Sites must follow manufacturer's guidelines for detection point placement. For example, if detector is to be placed in an exhaust duct, verify that the air velocities within the duct will not exceed the manufacturer's recommendation. The monitoring point should be located a minimum of three duct diameters downstream from an enclosure (ideally six to ten duct diameters) to ensure adequate mixing of the target gas with exhaust air. In addition, for the exhaust ducts with manifold design, the detector must be placed in the individual exhaust branch instead of the common one where multiple exhaust ducts merge into.

Figure 4: Gas Detector location



11.6.12 Sampling conditions at the area being monitored shall be evaluated to ensure that sampling points can effectively monitor the area. The evaluation should include, but is not limited to, the following: whether sample tubes need to be insulated, effectiveness of samples, gas aggressiveness, and sample time.

11.6.13 A Pitot tube sensor assembly, sized to fit the ductwork, should be located and installed as close to the source as possible to optimize detection sensitivity.

11.6.14 Chemcassette, Fourier Transform Infrared Spectroscopy (FTIR), electrochemical, electro-catalytic and Ultraviolet-Infrared (UVIR) are all acceptable technologies to monitor TGM.

11.6.15 The use of detector for monitoring a specific gas depends on the nature and characteristic curve of the gas. The detector types recommended for monitoring the different gasses in Micron:

No	Gas Type	Detector Technology
1	10%B2H6/H2	Chemcassette
2	5%82H6/N2	
3	0.5%BCL3/He	
4	BCL3	
5	CL2	
6	CLF3	
7	DCS	
8	20%F2/N2	
9	F2/Ar/Ne	
10	F2/Kr/Ne	
11	HBR	
12	HCD	
13	HCL	
14	NO	
15	0.5%PH3/N2	
16	5%PH3/N2	
17	5%PH3/He	
18	SICL4	
19	SIH4	
20	TMS	
21	WF6	
22	PH350ppm/N2	

No	Gas Type	Detector Technology
23	30%C2H4/He	Electro-Catalytic
24	C2H2	
25	CO	
26	C3H6	
27	CH4	
28	H2	
29	Natural Gas	
30	3.8%H2/N2	
31	O2	Electro-Chemical
32	Si2H6	
33	SO2	
34	HCL	
35	C4F6	Electro-Chemical + Pyrolysis or FTIR
36	CH3F	
37	COS	
38	NH3	Chemcassette
39	10%B2H6/H2	
40	5%B2H6/N2	
41	NF3	
42	Pyrophoric Organometallic liquid fire	UVIR
43	Silane Flame	
44	H2 Flame	
45	HC Flame	

11.6.16 A sample time of 60 seconds or less per point should be considered in the design.

11.6.17 TGM sensing equipment should be located in a centralized location to ensure quick response and repair due to failure.

11.6.18 Sample exhaust from TGM equipment shall be connected back to the same exhaust system.

11.6.19 Sensors shall be located between dampers and cabinets, positioned to ensure optimal results based on exhaust duct configuration.

11.7 Cables

11.7.1 All cables shall be low smoke zero halogen (LSZH) flame retardant type and shall be of adequate cross sectional area.

11.7.2 Lightning surge protectors shall be provided for all signals sent by copper cables external to the building, with the following requirements:

i. Provides protection from voltage spikes, sags, and surges.
ii. Provides minimum 40 dB reduction of electrical noise for frequencies from 50kHz to 50 MHz.
iii. Complies with UL 1449 for surge withstand capability.

11.7.3 Where cables and conductors serve as communication links between controllers, provide surge-protection circuits at each end which meet IEEE Standard C62.37 surge withstand capability tests.

11.7.4 Wire and Cable products.

i. Instrumentation cable 300V rated for 480V MCC/VFD applications, single pair, overall shield, type PLTC:
   - Individual Conductors: 18 AWG, stranded, tinned copper, polyethylene or PVC insulated, rated 105 degrees C, color coding per client standard.
   - Assembly: Twisted single pair, continuous 100 percent coverage aluminum-polyester shield, 18 AWG, stranded, tinned copper drain wire, overall PVC jacket. Maximum outer diameter 0.28 inches.

ii. Instrumentation cable 300V rated for 480V MCC/VFD applications, multiple pairs, overall shield, type PLTC:
   - Individual Conductors: 18 AWG, stranded, tinned copper, polyethylene or PVC insulated, rated 105 degrees C, black and white numerically printed and coded pairs.
   - Assembly: Individual twisted pairs having 100 percent coverage aluminum-polyester shield and 20 AWG, stranded, tinned copper drain wire. Provide conductor bundle with 100 percent coverage overall aluminum-

polyester shield and 18 AWG drain wire. Isolate group shields from each other. Wrap bundle with cable tape and cover with PVC jacket with overall PVC jacket.

iii. Ethernet Communications Cable: Industrial Ethernet ENET IP 10GBPS cabling: Category 6A UTP. Terminations: Shielded jack modules RJ45 IP20.
iv. Profibus DP Communications: Belden 3079A or Siemens equivalent, TSP 22 AWG Solid Bare Copper Conductors, Beldfoil (100% Coverage) + TC Braid Shield (65% Coverage), Foamed High-density Polyethylene Insulation, PVC Jacket, 300V 75°C
v. Fiber Optics Cable : 62.5/125 or 100/140 Multimode with ST connectors.

11.7.5 Cable and Wire Installation:

i. Install wires only in approved and complete raceways.
ii. Prevent mechanical damage to conductors during installation.
iii. Install and terminate all wire in accordance with manufacturer’s recommendations.
iv. Neatly arrange and lace conductors in all panelboards, pull boxes, and terminal cabinets by means of wire ties. Do not lace wires in gutter or panel channel. Install all wire ties with a flush cutting wire toe installation tool, use a tool with an adjustable tension setting. Do not leave sharp edges on wire ties.
v. Terminate solid conductors at equipment terminal screws with the conductor tightly wound around the screw so that it does not protrude beyond the screw head:
   - Wrap the conductor clockwise so that the wire loop is closed as the loop is tightened.
   - Do not use crimp lugs on solid wire.
vi. Thoroughly clean wires before installing lugs and connectors.
vii. Terminate stranded conductors on equipment box lugs such that all conductor strands are confined within the lug. Use ring type lugs if box lugs are not available on the equipment.
viii. Spare conductors are to be coiled and taped and mark as “SPARE”.
ix. Install instrumentation class cables in separate raceways from power cables.
x. Do not make intermediate terminations.
xi. All new wiring shall be tested for integrity and functionality after it is installed.

11.8 Terminal Blocks

11.8.1 Provide minimum of 10% spare terminals; identify spare terminals.

11.8.2 Space terminal strips so that future expansion, rework / maintenance, and field connections are accomplished without special tools.

11.8.3 Sectional Plastic Blocks:

i. DIN rail mounting with side entry.
ii. Rated for 300 or 600 volts suitable the application.
iii. Double sided.
iv. Capable of accepting wire sizes up to AWG 12.
v. Permanent, legible identification.
vi. Include end clamps and cover plates.
vii. Provisions for ganged terminal jumpers and test disconnect jumpers.

11.8.4 Elevate terminal strips so the top of the smallest terminal on the strip is even with the top of the adjacent wireway.

11.8.5 Install no more than two wires and a prefabricated jumper in any terminal block.

11.8.6 Provide fused terminal blocks with fuses for any loop that is powered from the panel (e.g., analog I/O, non-isolated Digital outputs). Provide knife disconnect terminals for digital inputs and isolated digital outputs.

11.8.7 Provide grounded shield terminals for analog I/O field connections.

11.8.8 Acceptable manufacturers:

i. Weidmuller
ii. Wago
iii. Or approved equal.

11.9 Cable Tray / Trunking

11.9.1 Signal wires and sample tubing should not be installed in the same raceway.

11.9.2 Control cables should be routed in ELV cable tray / trunking.

11.9.3 Conduit type to be used is depending on area:

i. Rigid metal conduit and fittings (for outdoor/indoor, concealed/exposed, embedded in Concrete, Underground, EMI sensitive area, :

- Rigid galvanized steel.
- Rigid aluminum conduit.
- Rigid steel conduit with external 40-mil PVC coating and internal phenolic coating over a galvanized surface.
- Fitting and conduit bodies: threaded type,

steel/aluminum.

- Hazardous area sealing fittings: UL-listed threaded - type conduit seal fittings and sealing compound suitable for hazardous location installations in accordance with NEC 501-5.

ii. Intermediate Metal conduit (IMC): galvanized steel and use fittings and conduit bodies specified above for rigid steel conduit. Used for outdoor/indoor, concealed, embedded in concrete, underground,

iii. Electrical Metallic Tubing: Galvanized tubing. Used for outdoor/indoor, concealed.

iv. Liquid-tight flexible conduit: Flexible metal conduit with PVC jacket; fittings and conduit bodies: metal with PVC jacket. Used for final connection where flexible connection is required to minimize vibration or to facilitate equipment/instrument removal or adjustment. Maximum length is 1 meter.

v. Rigid nonmetallic conduit: Schedule 40 PVC; fittings and conduit bodies: PVC. Used for outdoor/indoor, concealed, embedded in concrete, embedded in concrete, underground, corrosive nonhazardous area.

vi. Liquid-tight nonmetallic flexible conduit: plastic; fittings and conduit bodies: PVC. Used for final connection where flexible connection is required to minimize vibration or to facilitate equipment/instrument removal or adjustment. Maximum length is 1 meter.

11.9.4 Furnish and install new conduit and/or wire-way as shown diagrammatically on the installation plan drawings. Provide all supports, brackets and mounting hardware as required.

11.9.5 Control cables tray shall be separated from power cable tray.

11.9.6 Accessories and Fittings: Manufacturer’s standard clamps, hangers, brackets, splice plates, reducer plates, blind ends, barrier strips, connectors, and grounding straps.

11.9.7 Use bonding jumpers across expansion connectors for cable tray ground continuity. Provide bonding continuity between metallic cable tray sections and fittings in accordance with manufacturer’s instructions. Make connections to aluminum tray and fittings using an antioxidant compound. Bond cable tray systems to the power system’s ground every 50 feet or based on local code requirement. Do not use bare copper grounding conductors in aluminum cable trays.

11.9.8 Furnish and install approved Fire-stopping at all penetrations through rated fire walls/ceilings.

11.10 Power Requirement

11.10.1 Each control panel should be supplied dual power source: normal and emergency-backed UPS; or dual emergency-backed UPS from different source (different transformers).

11.10.2 Each incoming power to control panel is monitored with alarm enable.

11.11 Software

11.11.1 SCADA

i. The SCADA system shall guarantee that the TGMS continuously monitor the gas concentration in the environment and activate alarm devices and generate shutdown and interlock signals when the prescribed limits are exceeded.
ii. The SCADA server shall be redundant and high availability configuration.
iii. The SCADA shall have the following basic features:
    - Operating system (Standard Micron Windows deployment)
    - Redundancy.
    - Remote access.
    - Data acquisition and database configuration.
    - Network and data communications.
    - Supervisory controls.
    - Manual data entry.
    - Alarm/event processing.
    - Database facilities.
    - Graphic display and editor facilities.
    - System security.
    - System access shall be protected by at least two distinct components such as unique User ID and password.
    - Real-time database access module.
    - Report generation.
    - Historical trending, data collection, and retrieval via Historian.
    - Real-time trending.
    - Auto logout.
    - System shall provide automatic log off after a predefined period of non-attendance to prevent

unauthorized access.

- Alarm/event to paging system.
- On-line tutorial.
- OPC/UA connectivity.
- Support SQL.

iv. The alarm limits in the TGMS are variable. These alarm limit parameters may only be changed via audit trail by an authorized person protected by adequate access control and system privileges. All changes are logged.
v. Alarm indication and handling shall be provided. Alarms will be logged in an audit trail. Alarm ranges will be configurable. Sorting and filtering of alarms will be possible.
vi. The TGMS shall provide authorized system access for predefined user groups like System Administrator, Supervisor, and User. Individual system functions will be assigned or limited to different user groups.
vii. Creation, modification, and deletion of monitoring parameter settings, alarm limits, shall be restricted to authorized personnel and shall be controlled.
viii. Modification and deletion of monitored raw data must be prevented. Monitoring raw data must be protected.
ix. Historical Data shall be retained for minimum 2 years.

11.11.2 Third Party Interface

i. All detection measurements shall be recorded in FMCS SCADA system.
ii. OPC server shall be provided in the TGMS SCADA.
iii. Reading data between TGMS and FMCS/Third party should be on demand and only out of dedicated memories.

11.11.3 Cyber Security

The system shall comply with Micron cyber security policy. Other additional cyber protection enhancements will be described below for recommendation:

i. All controllers should reside in locked panels/cabinets
ii. Programing and Debugging Tools connected to isolated networks only.
iii. Disconnect Programming and Debugging tools during normal operation.
iv. Block unused communication interface ports at network switches.

11.11.4 IT Requirement

i. The SCADA server and workstation will be coordinated with IT for integration.

ii. Control interlock and monitoring should be dedicated and separate network.

11.12 Commissioning

11.12.1 General Requirement

i. Systematic process of ensuring systems perform interactively per Project design intent and Owner's performance requirements.
ii. The system is to be tested in accordance with manufacturer's recommendations and NFPA or equivalent.
iii. Inspection and Test Plan should be submitted together with Quality management system for user's approval during design engineering stage.
iv. The organizing, coordination of personnel, and scheduling of all tests to be the responsibility of the Vendor. Notification of testing to be submitted to each testing participants a minimum of 2 weeks prior to the requested testing date.
v. Test programs and procedures to be created by the Vendor. Test procedures to have signoff spaces for the Owner and the Vendor. Where Owner-furnished test procedures exist, they are to be used in place of Vendor's test procedures.
vi. Manufacturer's service engineers or technicians to be present for pretests and Owner acceptance tests.
vii. Pretest the system for satisfactory operation prior to Owner system acceptance testing. Prior to the Owner witnessing any testing, a system printout demonstrating satisfactory pretest results to be submitted to the Owner 2 days prior to the anticipated test date.

11.13 FAT (Factory Acceptance Test)

11.13.1 Factory Acceptance Test (FAT) requirements:

i. The equipment will be inspected during manufacturing and before shipment for site installation. Furthermore, Micron reserves the right to perform formal shop inspections. Vendor shall advice Micron of the fabrication status on a monthly basis. Prerequisite for shipping is the Micron approved FAT.
ii. The Vendor prepares the FAT procedures for approval of Micron.
iii. The FAT will consist of the following checks:
   - Check conformance to the Technical Specifications of Micron and the specifications of the Vendor.

- Check for completeness.
- Check for instrument materials.
- Verify correctness of nameplate data.
- Verify all SIL certificate, IS certificate & explosion certificate etc. for all the material to be supplied.
- If possible – simulation of operation.
- Verify interfaces connection ports to be installed.
- Verification of redundancy of devices and network.
- Verification of device failure scenarios for system robustness.
- Verification of fail-safe scenarios.
- Verification that all alarms and interlocks work as specified.
- 100% test of point and function.
- Review of the complete specified documentation package.
- Review of equipment conformance with relevant codes and guidelines.

iv. All Vendor’s internal testing before the FAT should be documented properly according to the Vendors internal procedures. Tests reports should be available before Micron’s arrival at Vendor shop for FAT.
v. The tests shall be carried out at Vendor’s expense.
vi. IF the system does not meet and perform in accordance with Vendor warranty obligation, Vendor will, at its expense, implement all the necessary modifications and afterwards the tests shall be repeated.
vii. After Micron approval, the system will be prepared and shipped to the site.
viii. The complete set of internal testing documents must be available before FAT. One copy shall be sent to Micron before FAT. No FAT shall be carried out without documents availability.

11.14 Installation Inspection

11.14.1 Once the control panel and relevant field devices have been totally installed and connected, loop checking activities have been completed, and the relevant reports have been approved by Micron, the TGMS commissioning can be performed.

11.14.2 The panel passes a visual inspection and complies with the following:

i. Outer enclosure is labeled appropriately with ID tag.
ii. Devices and modules are installed per the drawings.
iii. Wiring is gathered with tie-wraps or bundled within wire-mold channels.
iv. Wiring, devices and modules are labeled within the panel in accordance with the drawings.
v. Wiring terminations are adequate and firm, and wiring utilizes the color code established within the drawings.
vi. Wiring cable shields are terminated at one end to power supply ground as appropriate.
vii. Confirm panel, equipment rack and camera poles are grounded to appropriate grounding source.

11.14.3 Devices are mechanically fastened properly, securely and firmly on mounting supports.

11.14.4 Field cable and sensor installation shall be visually inspected as per specification and I/O list.

11.15 SAT (Site Acceptance Test)

11.15.1 The final acceptance at site will be carried out after mechanical completion of the system at Micron facility. The purpose of the SAT is to verify that the system is supplied and installed and that it operates according to the relevant specification.

11.15.2 At the completion of installation of panels and field devices, each system to be tested by the Vendor with a factory-trained and certified field technician to show the following:

i. Panel Verification Testing: At the completion of panel assembly, each TGMS panel to be tested by a factory-trained and certified field technician to show the following:

- Jumper and dip switch settings are in accordance with the manufacturer’s requirements.
- Confirm input voltage (120Vac or 240Vac) is within appropriate tolerance range, and that the panel indicates normal operating condition upon the 120 or 240 Vac power application.
- The panel to accept downloaded program address and configuration information. The panel to show a normal standby condition upon completion of download, including that it is free from grounded or open circuits.
- The panel indicates a ground or open circuit occurs upon application of ground fault or open circuit.
- Confirm power supply output voltages are within

appropriate tolerance range and that the battery and charger indicate proper charge and operation where batteries and battery charging systems are used.

- Simulate power failure and test all components when system is operating on battery or UPS. Confirm no loss of functionality occurs.
- The communication and device operation within each panel, between each panel and the head-end controls, and among all panels and the head-end controls as an entire system is established and verified.
- The appropriate communication and alarm annunciation are demonstrated upon a test simulation program being loaded and executed.
- Defects to be corrected at once and the test re-conducted at no additional cost to the Owner until the defect as been corrected.

ii. Network Backbone Verification Testing: At the completion of assembly and/or procurement, but before final installation is completed, TGMS panels and backbone communication wiring to be tested by the Vendor with a factory-trained and certified field technician to show the following:

- The system is properly calibrated and adjusted in accordance with manufacturer’s recommendations.
- The system is free from grounded or open circuits.
- The control equipment will indicate when a ground or open circuit occurs.
- Contact-initiating devices are operating normally.
- Simulate power failure and test all components when system is operating on battery or UPS. Confirm no loss of functionality occurs.
- Where fiber optic cable is used, use an Optical Time Domain Reflectometer (OTDR) to document attenuation (optical power loss) in both directions from the ends of each fiber cable.
- Confirm all exterior-mounted TGMS ethernet network components are equipped with appropriate surge protection.
- Where ethernet networks are used for TGMS, document the following using network testing means such as PSPing or similar software:
  - Network bandwidth throughput – maximum rate at which frames can be transmitted from the

source to the destination with zero lost frames or errors.

- Network Latency – the total time it takes for a frame to travel from source to destination. The minimum, average, and maximum latency values for each frame size shall be reported.

- The communication and device operation within each panel, between each panel and the head-end controls, and among all panels and the-end controls as an entire system is established and verified.
- The appropriate communication and alarm annunciation are demonstrated upon a test simulation program being loaded and executed.
- Defects are to be corrected at once and the test re-conducted at no additional cost to the Owner until the defect has been corrected.

iii. Where internet protocol (IP) devices are used for TGMS, record the following:

- All network device IP addresses, MAC addresses, subnet masks and default gateway assignments.
- Record DNS hostname, domain and server assignments.
- Confirm default passwords have been changed on all devices and provide an electronic list of all changed device passwords.

iv. Correct operation and indication of grounds, shorts, and opens is tested at every point with the final operating software program.

v. Perform functional specification in the final acceptance test procedure that is based on specified control operation in this specification.

vi. The point-to-point test, functional test, and calibration must take place online from the sensor to the SCADA system.

- A calibration is a comparison of measuring equipment against a standard instrument of higher accuracy to detect, correlate, adjust, rectify and document the accuracy of the instrument being compared.
- Functional test shall be carried from sensor, PLC, and actuator/final element, with SCADA to monitor all the function.

vii. The final acceptance test to be witnessed by Owner, or an authorized representative of the Owner, the Vendor, and where appropriate, Authority.

viii. Defects to be corrected at once and the test re-conducted at no additional cost to the Owner until the defect is corrected.

11.15.3 The SAT testing also includes:

i. Check of completions of open items from FAT
ii. Check of transport damages.
iii. Check scope of supply.
iv. Proper operation of the systems.
v. Repetition of dedicated tests carried out at Vendor shop during FAT.
vi. Verification of final documentation (incl. change and revision control and traceability of documentation)

11.16 Documentation

11.16.1 Quality Management System should be in place and used for documentation of the commissioning activities.

11.16.2 The Vendor should provide a document list of his standard documentation and commissioning documentation package together with examples for review and final approval by Micron including a description of the vendor internal quality system.

11.16.3 Prior to the execution of any commissioning activity of the vendor, Micron is to be supplied with the respective protocols and test specification for approval – these must be made available to Micron at least 4 weeks before execution of the testing.

12.0 Document Control

12.1 **Approval**: GLOBAL_FAC_GFTT_APPR

12.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

12.3 Final Viewer Access

All Micron Team Members at all Sites

12.4 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures.

12.5 Review

Biennially (two years)

13.0 Revision History

Revision #	Section Changed	Details of Changes	Date
1	NA	New document	05/27/20
2	3.0 4.0 6.0 10.0 11.0	add AC, DC, BKM, CPU, EMO, ESTOP, EHS, I/O, GPS, PSM, PLC, P&ID, SIL, SCADA, HMI, MTBF Revise GFTT Lead responsibilities to: "Incorporate all relevant BKM, lesson learned and emerging technologies in System Standard document." From "Ensure Best Known Methods (BKM), lessons learned and emerging technologies are documented in System Standard" Remove words: "Best Known Method" from "Site Project Construction Teams" 6.1 Remove words: "Toxic Gas Monitoring System" 6.4 Remove words: "valve manifold boxes", "building isolation cabinets", "gas distribution boxes", "field distribution boxes", "exhausted valve boxes" 6.5 Remove words: "Failure Mode and Effects Analysis", "Hazard and Operability" 6.6.1 Remove words: "Environmental, Health, and Safety", 'Process Safety" 10.12 Remove words: "Emergency Stop" 11.1.2 Remove words: "Programmable Logic Controller", "input/output" 11.1.6 Remove words: "Human Machine Interfaces" 11.6.11 Figure 3 is corrected to Figure 4. 11.6.15 CH3F detection type changed to "electrochemical + pyrolysis or FTIR" from "FTIR";Remove"CH2F2"	05/27/22

Appendix A: Deviation List

Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall be the following:

Site: ______________________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT